

Campus Resource Management

Campus Reserve is a browser prototype for finding campus spaces, requesting reservations and tracking permission decisions.

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Main features and use cases

Shared availability and local notifications connect the booking workflow.

Student / society

Browse room availability, submit a reservation, track its status and cancel when needed.

Permission in-charge

Inspect pending requests and record approval or rejection with review details.

Campus administrator

Add or update room and society records through dedicated management screens.

Technology stack

Interface and tooling

React 19 and TypeScript 6

Vite 8 and React Router 7

Tailwind CSS 4 and Lucide React

Data and application logic

Browser localStorage stores demo records.

A TypeScript service manages bookings, overlap checks and notifications.

Current scope: frontend prototype with demo personas and no backend database.

Prototype workflow and architecture

Browser architecture

React screens use typed application logic and a storage service to read and write localStorage.

All booking data stays in the same browser.

1. Select

Choose a room, date and available time slot.

2. Request

Complete the eight-step form and review details.

3. Validate

Check timing, conflicts and night permission.

4. Review

Save as Pending, then approve or reject and notify.

Conclusion and future scope

The prototype demonstrates room discovery, request submission and review.

A campus-wide service needs the additions below.

Verified identity

Institutional sign-in and server-side authorization.

Shared database

Synchronize bookings across users and devices.

Reliable booking rules

Atomic conflict checks and a decision audit trail.

Campus validation

User testing and targeted notification delivery.